# Nikhil Gangaram

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## Education

**Cornell Tech** — New York City, NY
*Masters of Engineering in Computer Science* — *Aug 2025 – Dec 2025*
- Dropped out Late First Semester : Trustworthy AI, Interactive Device Design

**Worcester Polytechnic Institute** — Worcester, MA
*Bachelor of Science in Robotics Engineering & Computer Science Minor* — *Aug. 2022 – May 2025*
- GPA - 4.0 : Robotic SLAM, Robotic Manipulation, Quantum Information

## Experience

**AI Researcher** — Present
*Distyl AI* — *San Francisco, CA*
- Working on cross-modal model steering and behavioral alignment.

**Robotics Software Consultant** — Jan. 2025 – Aug. 2025
- Developed a computer vision system to estimate plant health in vertical farming @ Untill
- Developing a task-graph based abstraction layer for general robotics software @ Cerulion

**Visiting Researcher** — Oct. 2024 – Dec. 2024
*Kyoto University of Advanced Science (KUAS)* — *Kyoto, Japan*
- Worked with Prof. Ryosuke Matsumoto to develop Equivariant Graph Neural Network based interatomic potentials which predict the effects of hydrogen vacancies to mitigate embrittlement in magnesium alloys

**Research Intern** — June 2024 – Oct. 2024
*MIT Lincoln Laboratory* — *Lexington, MA*
- Worked with Luis Alvarez to deploy multi-aircraft systems which utilize the Soft Actor-Critic architecture to protect civilians in the case of failure and provide aid during natural disasters
- Work was accepted into AIAA ML/AI in Air Transportation : "Part II Risk Reduction to Populated Areas"

**Student Researcher** — Sep. 2023 – Present
*Worcester Polytechnic Institute (WPI)* — *Worcester, MA*
- Working with Prof. Carlo Pinciroli to build a multi-agent toolbox for MathWorks, focusing on multi-robot task allocation and distributed SLAM for both simulation and real hardware
- Worked with Prof. Daniel Reichman on fine-tuning LLMs with NP-hardness reductions to enhance reasoning capabilities. Our work, The Karp Dataset, was published in the NeurIPS 2024 Workshop MATH-AI

**R&D Software Lead** — Aug. 2023 – Present
*WPI HPRC* — *Worcester, MA*
- Led the development of an Extended Kalman Filter and a Model Predictive Controller for onboard, real-time control of a model rocket. Currently building a simulator in Unreal Engine to extend with an LSTM

## Projects

**Experiential Robotics Project (XRP)** | *Python, Markdown, RST* — Ongoing
- Prototyped one of the first Socratic, LLM-based learning modules for classroom use. Our work, WIP: Integrating an LLM-powered Assistant into High School Robotics Education, was published in IEEE Frontiers in Education

**HURON** | *Python, MATLAB, ROS / Drake* — 2023
- Started the implementation of a Nonlinear Model Predictive Control (NPMC) algorithm in Python and MATLAB to realize dynamically stable locomotion. The bipedal robot was simulated using ROS and Drake

**Project Capricornus - 2022 WPI High Powered Rocketry Club (HPRC)** | *Lua, C++/C, Solidworks* — 2022
- Implemented scripts for an autonomous drone using Lua and ArduPilot
- Developed sensor libraries for weather-station cubes in embedded C
- Designed and fabricated an arm-folding mechanism for a cube-sat form factor drone in Solidworks